C1 Four Topology MOSbius

Four Topology MOSbius

Discord name	Affiliation	Experience	Role
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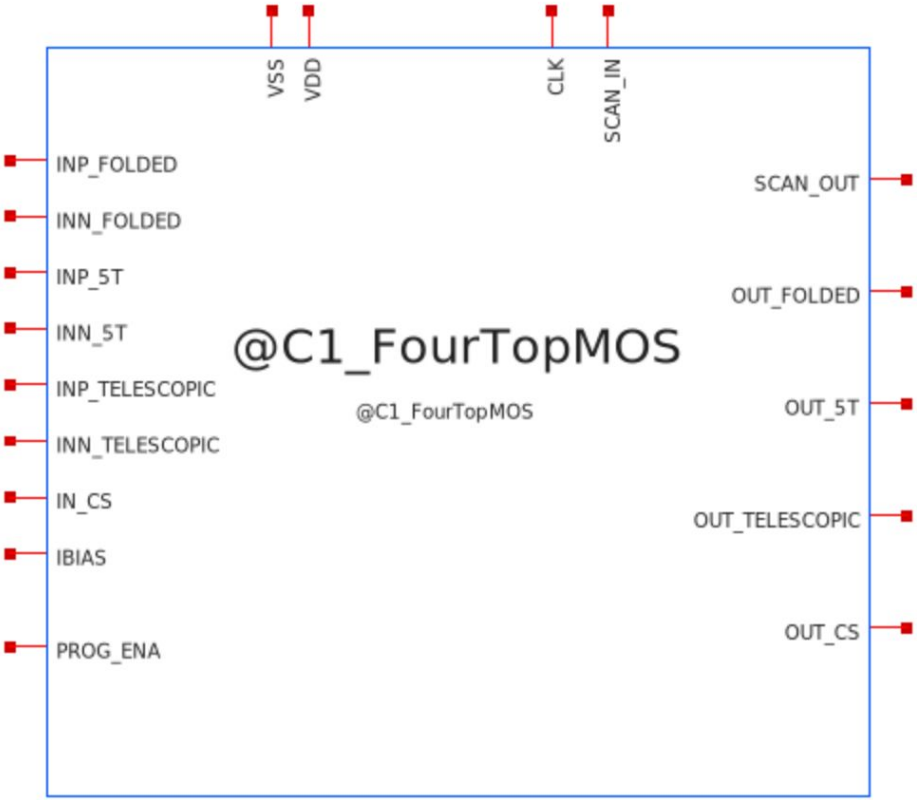
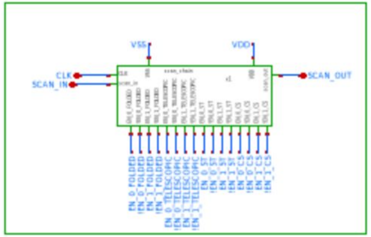
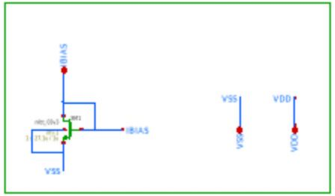
Manwell	Columbia University	Undergraduate	Team Member


VSS
VDD
CLK
SCAN_IN
INP_FOLDED
INN_FOLDED
INP_5T
INN_5T
INP_TELESCOPIC
INN_TELESCOPIC
IN_CS
IBIAS
PROG_ENA
SCAN_OUT
OUT_FOLDED
OUT_5T
OUT_TELESCOPIC
OUT_CS
@C1_FourTopMOS
@C1_FourTopMOS

SCAN CHAIN

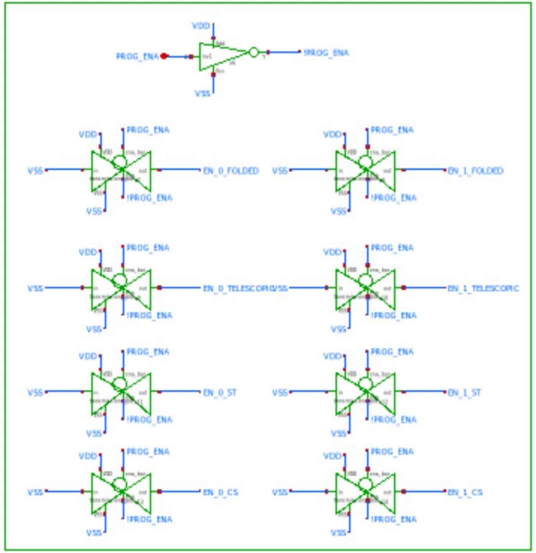


BIAS AND POWER PINS

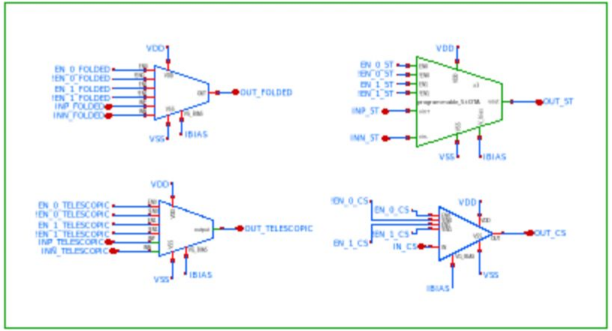


PROGRAMMING ENABLE

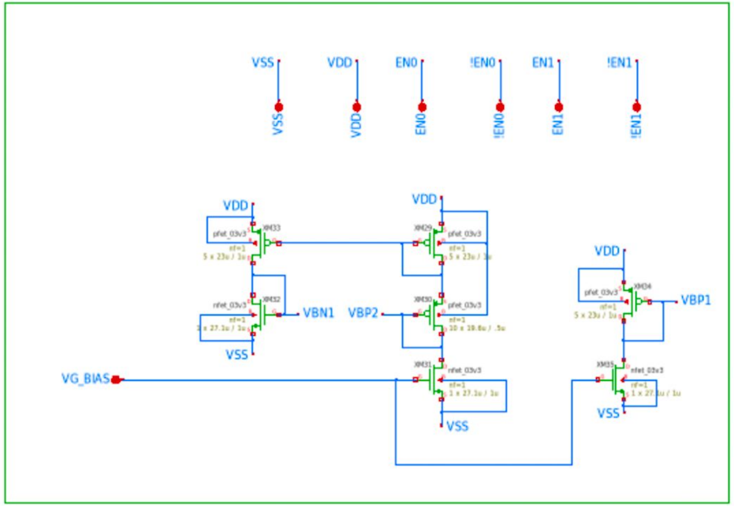
Pull Downs on Enable Pins

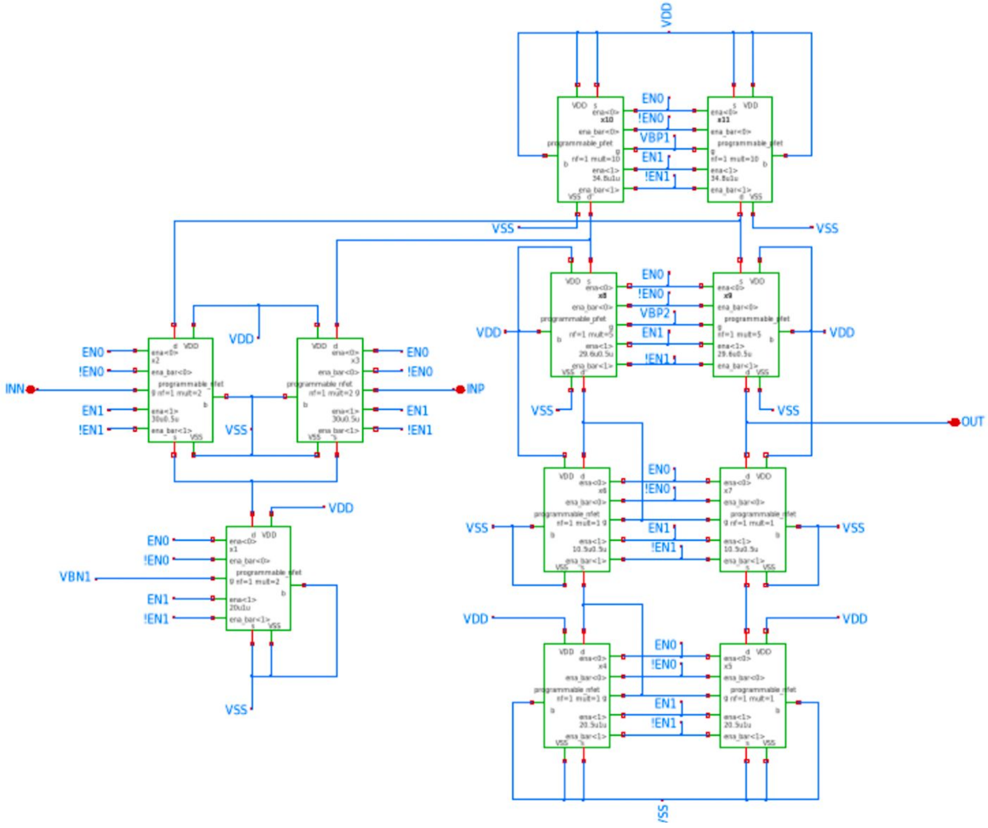


CORE AMPLIFIERS



Bias Voltage Generation and Pins

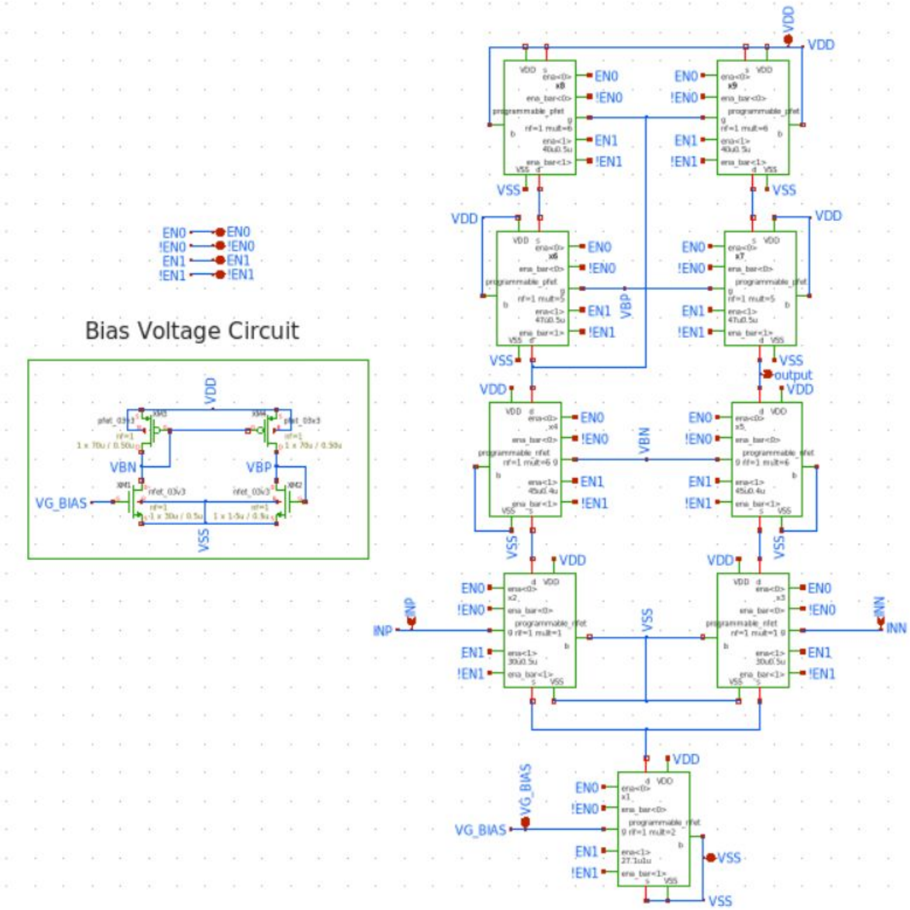




Folded Cascode Sizing

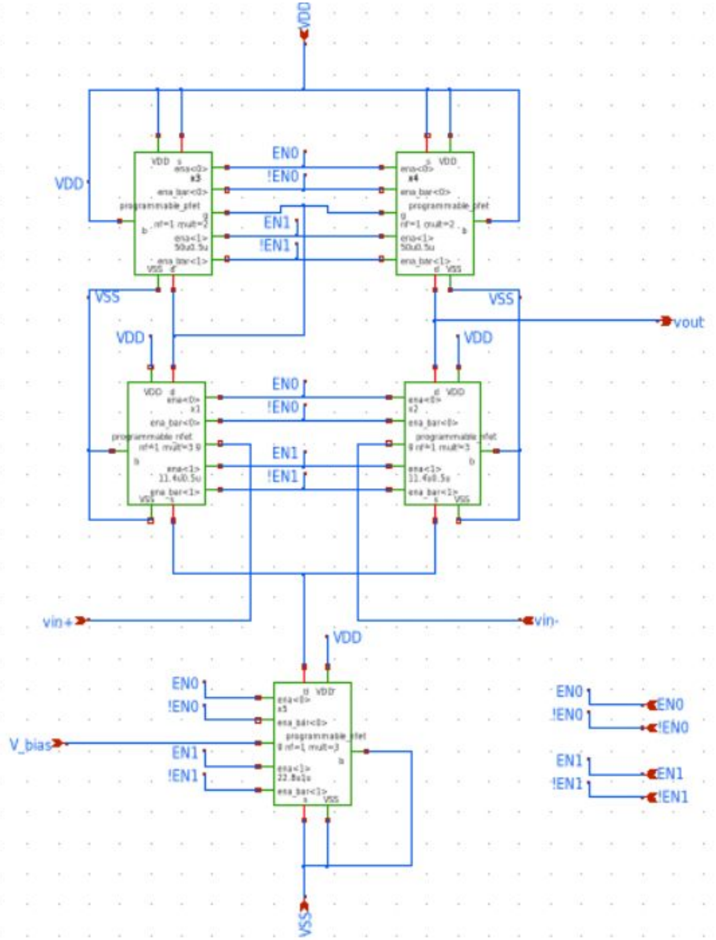
Device	Role	W (µm)	L (µm)	ID (µA)
XM10/XM11	NMOS input pair	60	0.5	75
XM12	NMOS tail	40	1	150
XM2/XM5	PMOS current sources	174	1	150
XM3/XM4	PMOS cascodes	148	0.5	75
XM6/XM7	NMOS cascodes	10.5	0.5	75
XM8/XM9	NMOS mirrors	20.5	1	75

- (Default sizing shown, for 2x, 3x, and 4x sizing, the width and bias current would scale respectively for all amplifiers)


Bias Voltage Circuit
EN0
!EN0
EN1
!EN1
VDD
VSS
VBN
VBP
VG_BIAS
INP
INN
output
programmable_pfet
programmable_nfet

Telescopic Cascode Sizing

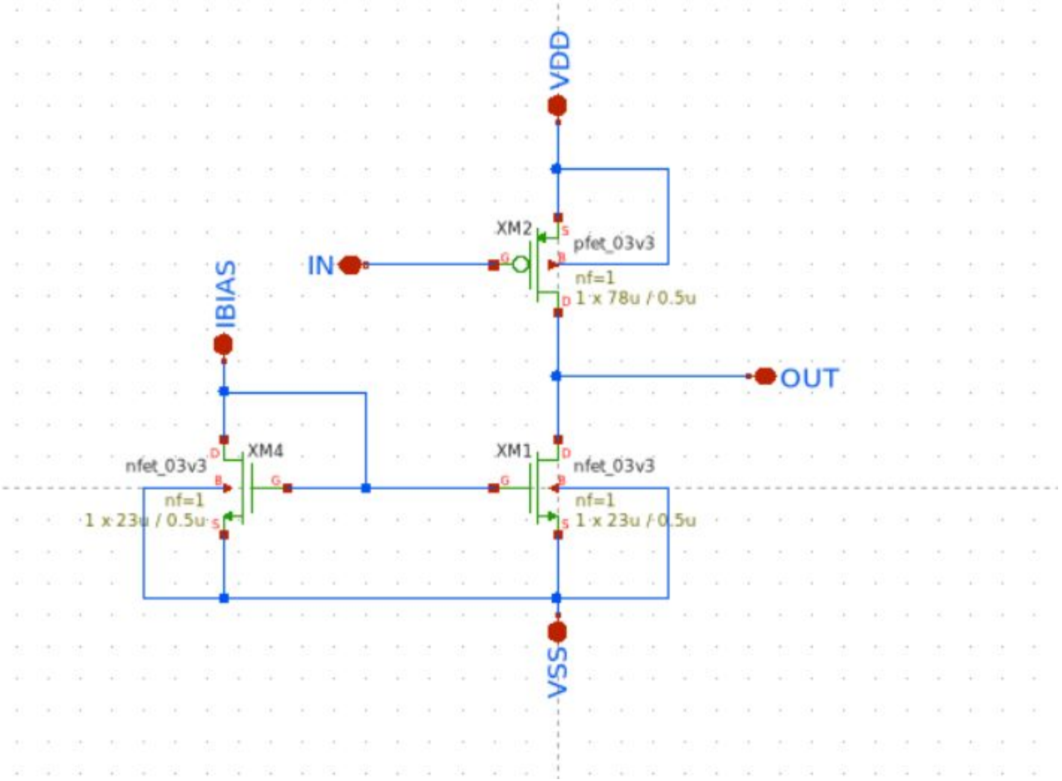
Device	Role	W (µm)	L (µm)	ID (µA)
XM1	NMOS Tail	60	0.5	200
XM2/XM3	Input Pair	30	0.5	100
XM4/XM5	NMOS Cascode	180	0.5	100
XM6/XM7	PMOS Sascodes	235	0.4	100
XM8/XM9	PMOS Current Sources	240	0.5	100


VDD
VDD
EN0
!EN0
EN1
!EN1
VSS
VSS
vout
VDD
VDD
EN0
!EN0
EN1
!EN1
vin+
vin-
VDD
EN0
!EN0
V_bias
EN1
!EN1
EN0
!EN0
EN1
!EN1
VSS

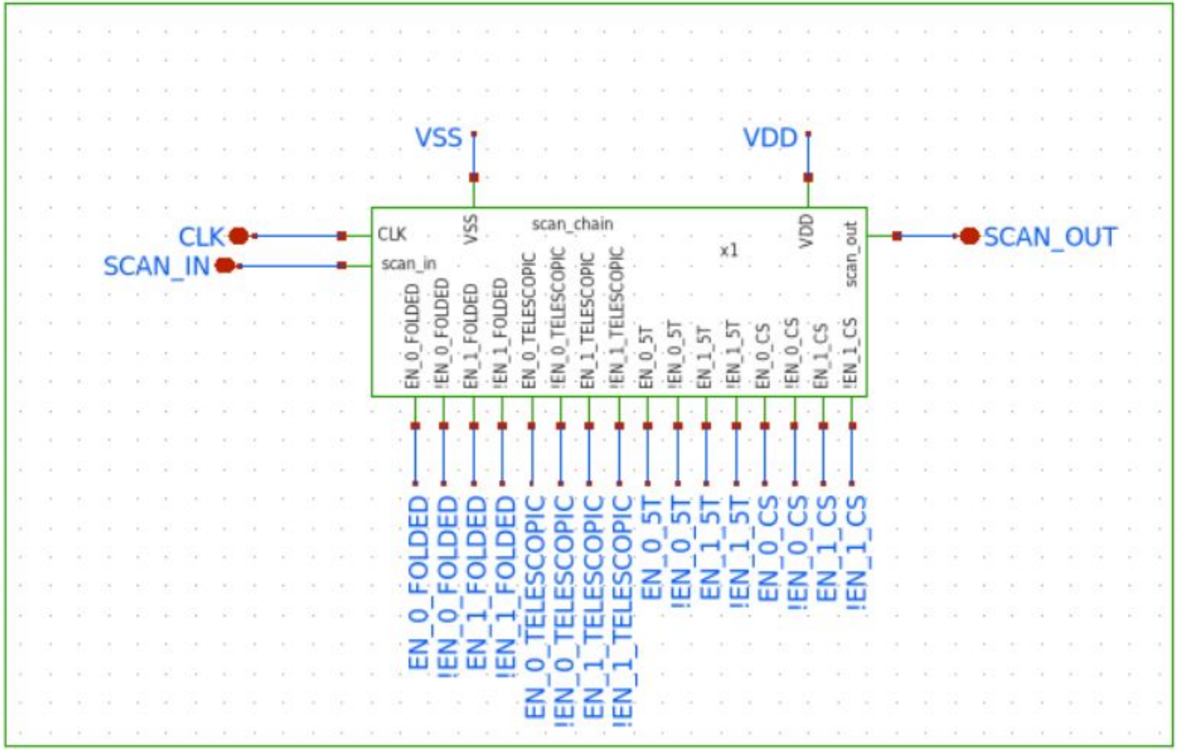
5-Transistor Sizing

Device	Role	W (µm)	L (µm)	ID (µA)
XM1/XM2	Input Pair	34.2	0.5	150
XM3/XM4	PMOS Active Load	100	0.5	150
XM5	NMOS Tail	68.4	0.5	300

Common-Source Amplifier

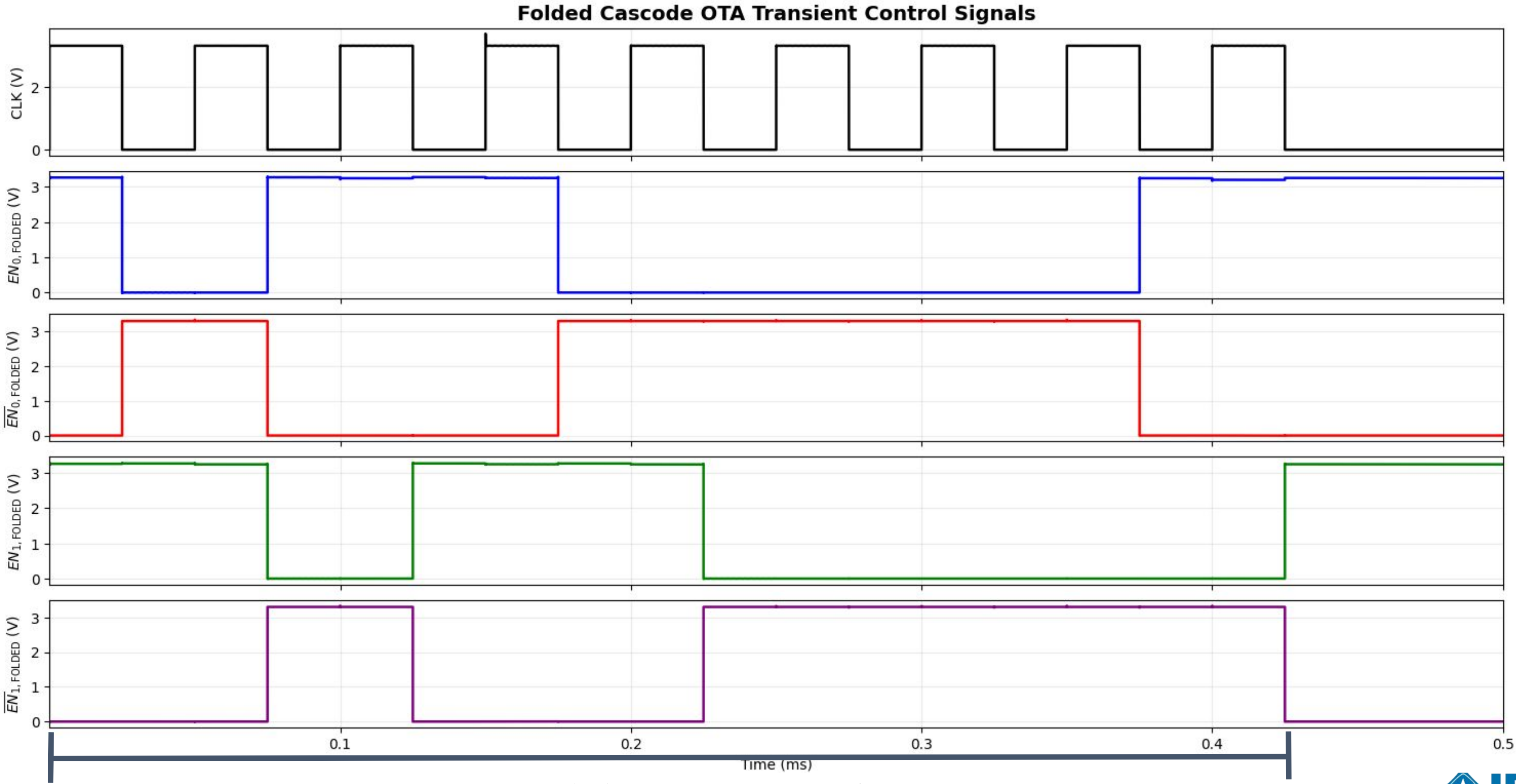


SCAN CHAIN


VSS
VDD
CLK
SCAN_IN
SCAN_OUT
scan_chain
x1
CLK
scan_in
VSS
VDD
scan_out
EN_0_FOLDED
!EN_0_FOLDED
EN_1_FOLDED
!EN_1_FOLDED
EN_0_TELESCOPIC
!EN_0_TELESCOPIC
EN_1_TELESCOPIC
!EN_1_TELESCOPIC
EN_0_5T
!EN_0_5T
EN_1_5T
!EN_1_5T
EN_0_CS
!EN_0_CS
EN_1_CS
!EN_1_CS

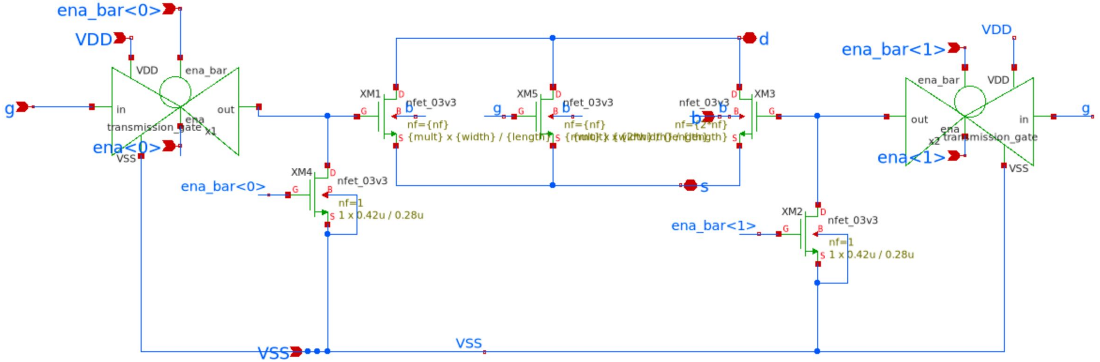
Clock and Scan-In Signal are to be generated by a Raspberry Pi running micropython (can also be replaced by any other microcontroller). Data for any possible configuration can be propagated in .425 ms, and verified via the Scan-Out Pin. In the next slide, the D Flip-Flop outputs of EN0 and EN1 (and compliments) are displayed, propagated in .425 ms. At .425 ms the clock turns off, and with the enable bits set in place until the user chooses to reconfigure the Four Topology MOSbius.

*During this data propagation, in transient simulations, you will notice voltage spikes do to digital switching, which disappear after the clock is turned off.

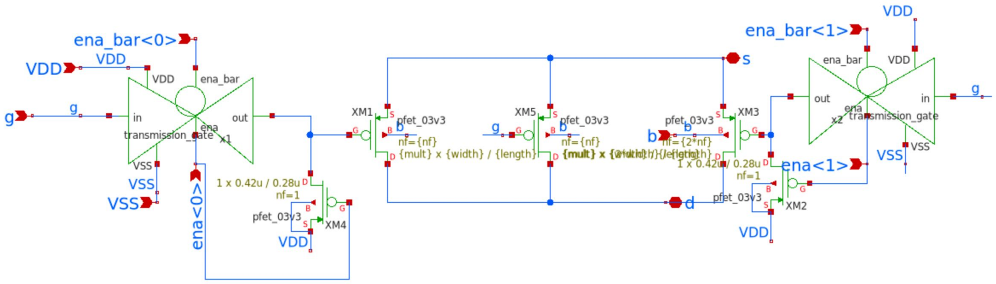


.425 ms Data Propagation

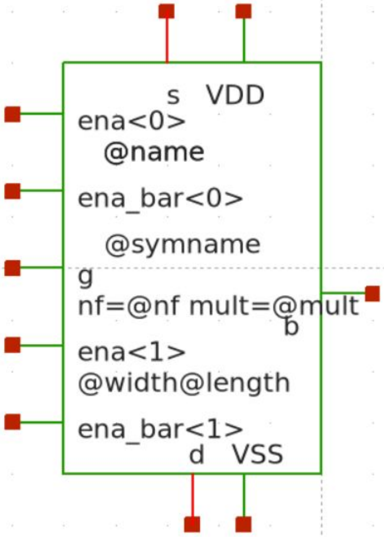
Programmable NFET



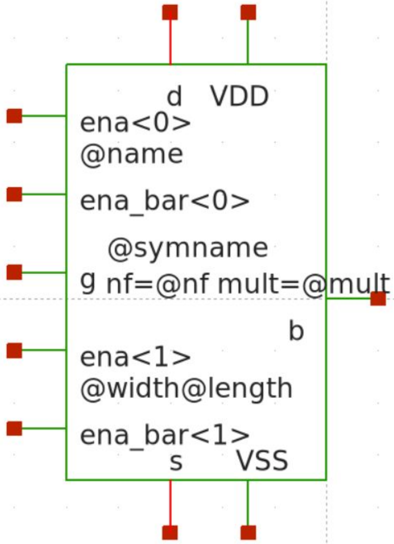
Programmable PFET



Programmable PFET

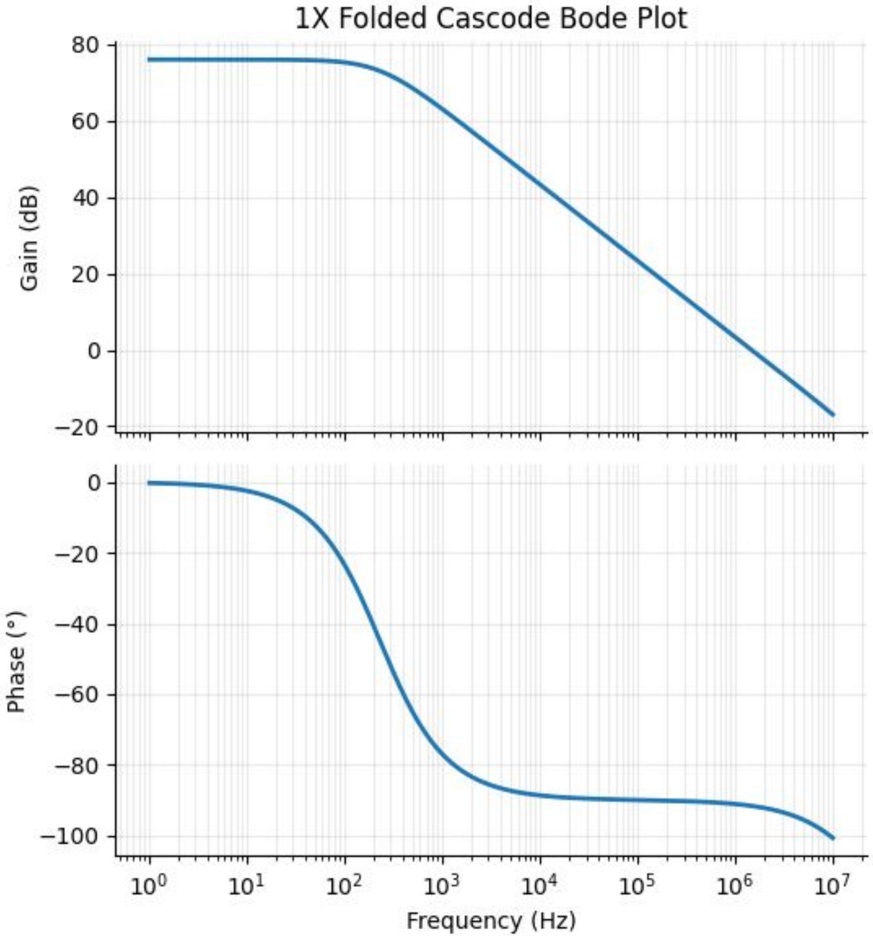


Programmable NFET

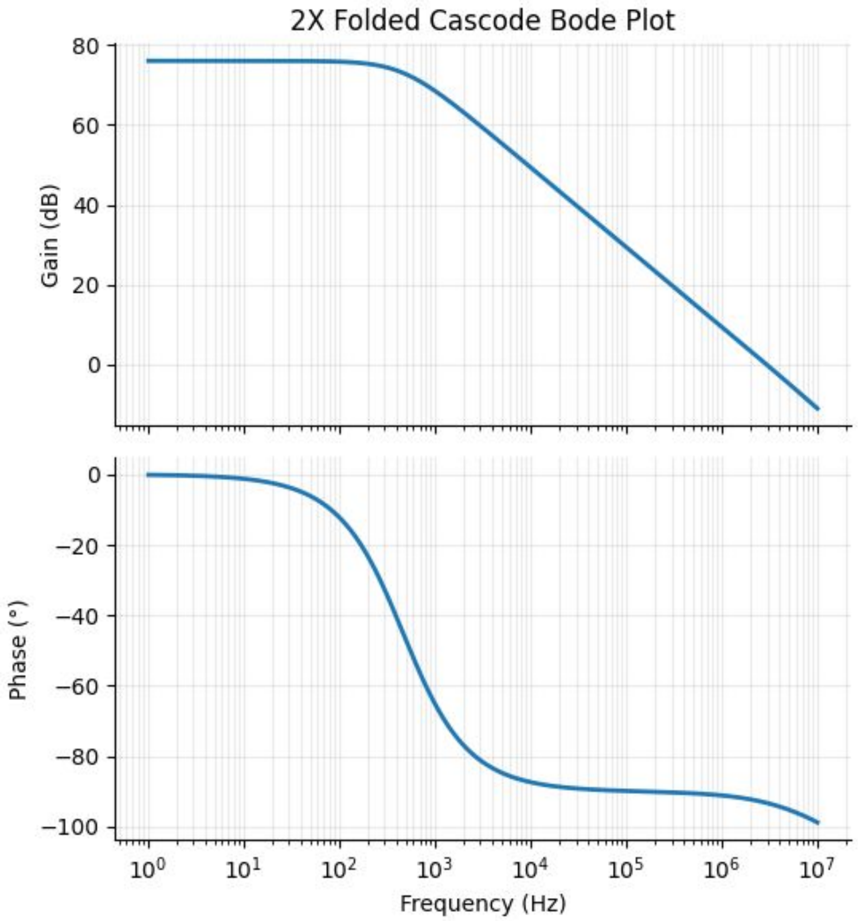


AC Simulations

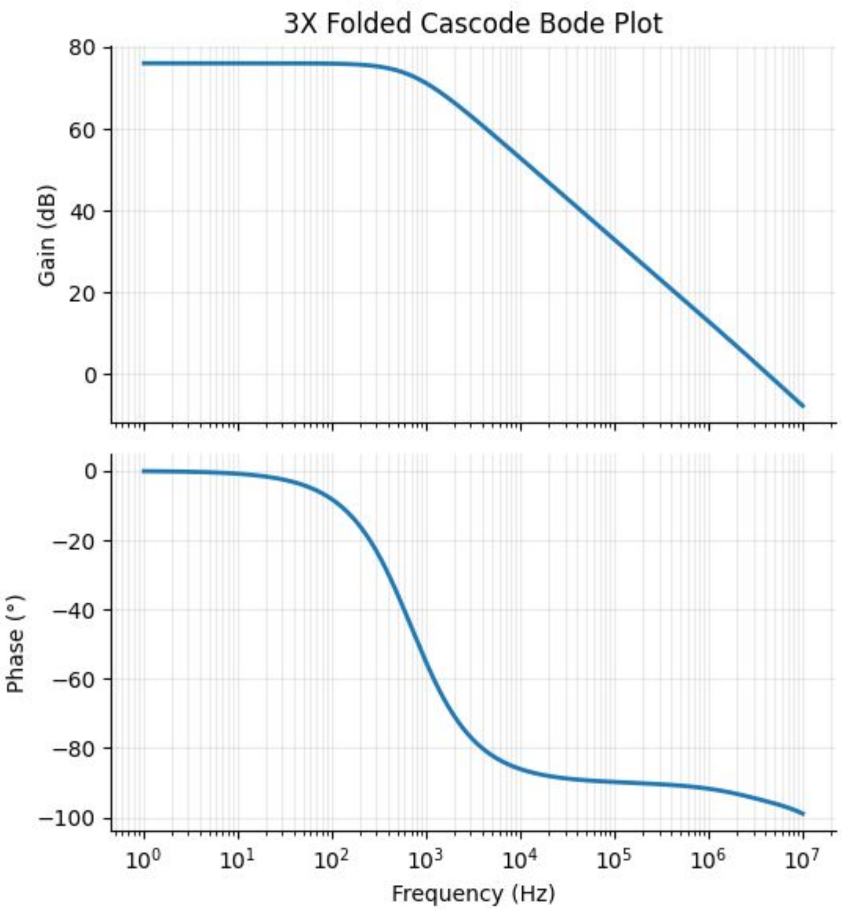
The following AC Sims were extracted by hard coding the Programming Transistors (Setting the Enable Pins high or low without the need of the Scan Chain). AC sims were unable to be done directly via the Top-Level as SPICE was unable to find an operation point due to the presence of the digital platform. Instead, GBW was extracted via the GBW product. There is an error margin as result were limited by 1uS resolution simulations. These results can be found in the last slide.



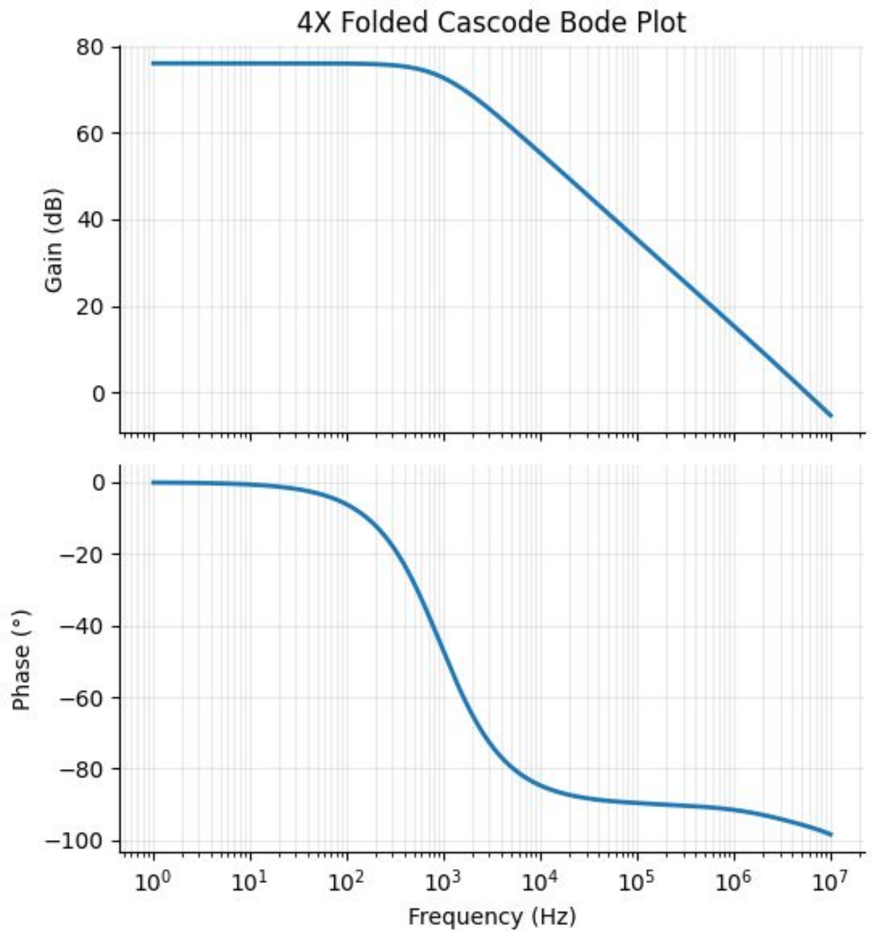
Gain = (76.04 dB)
PM = 88.3°
GBW = 1.51 MHz



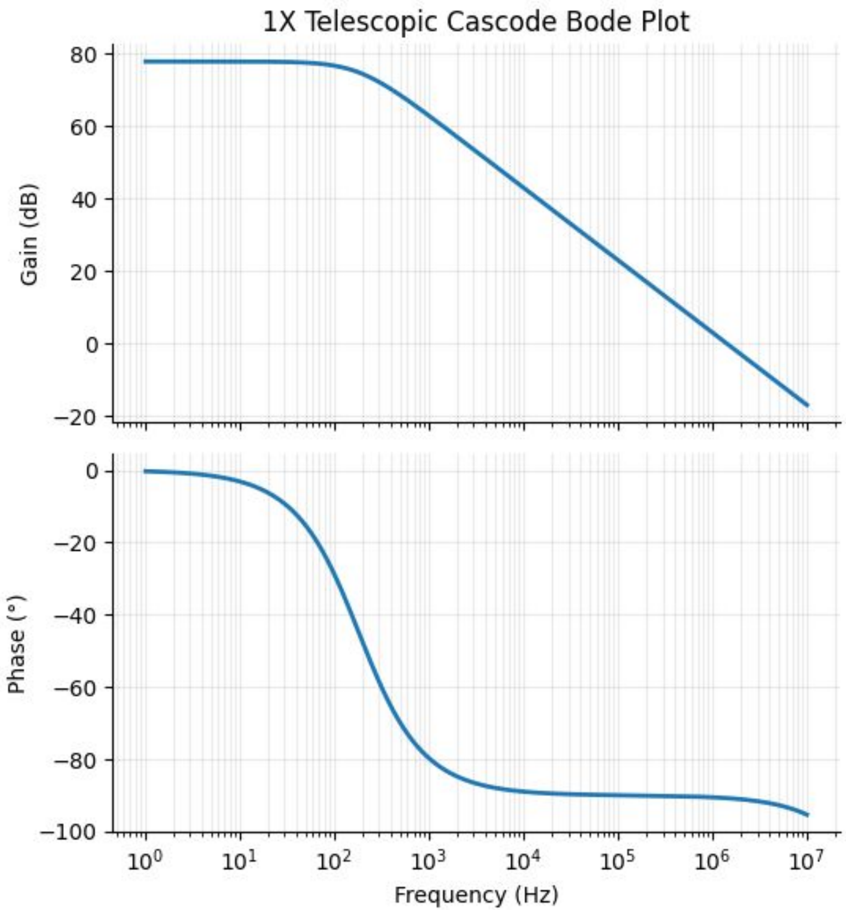
Gain = (76.04 dB)
PM = 86.8°
GBW = 2.95 MHz



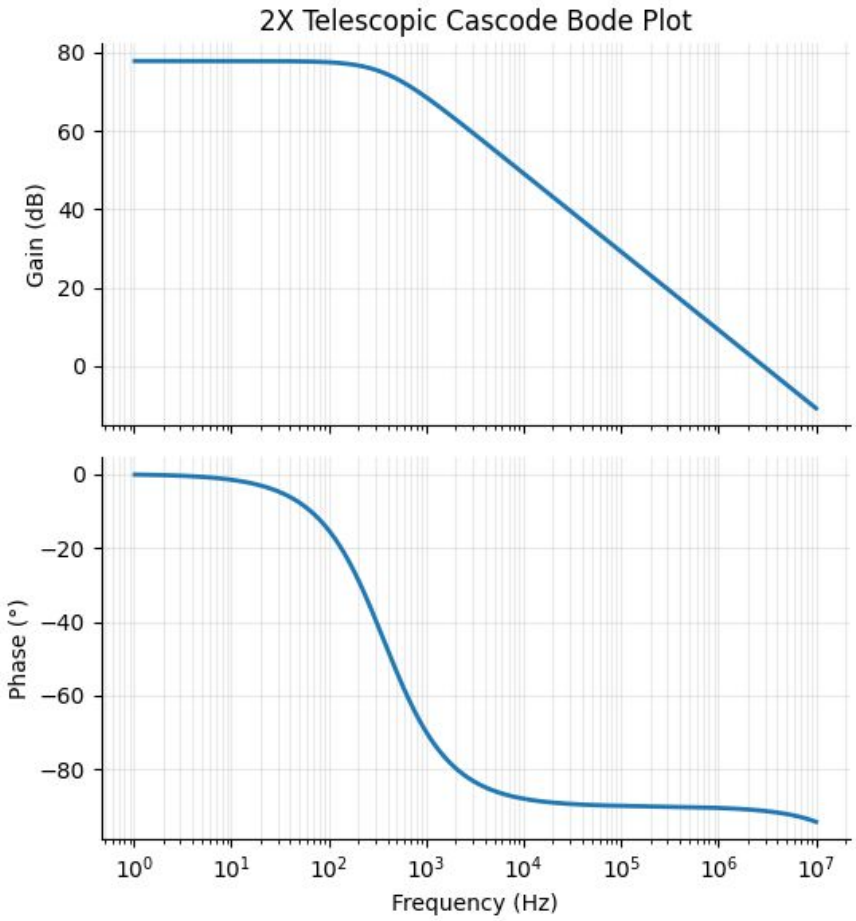
Gain = (76.04 dB)
PM = 84.6°
GBW = 4.27 MHz



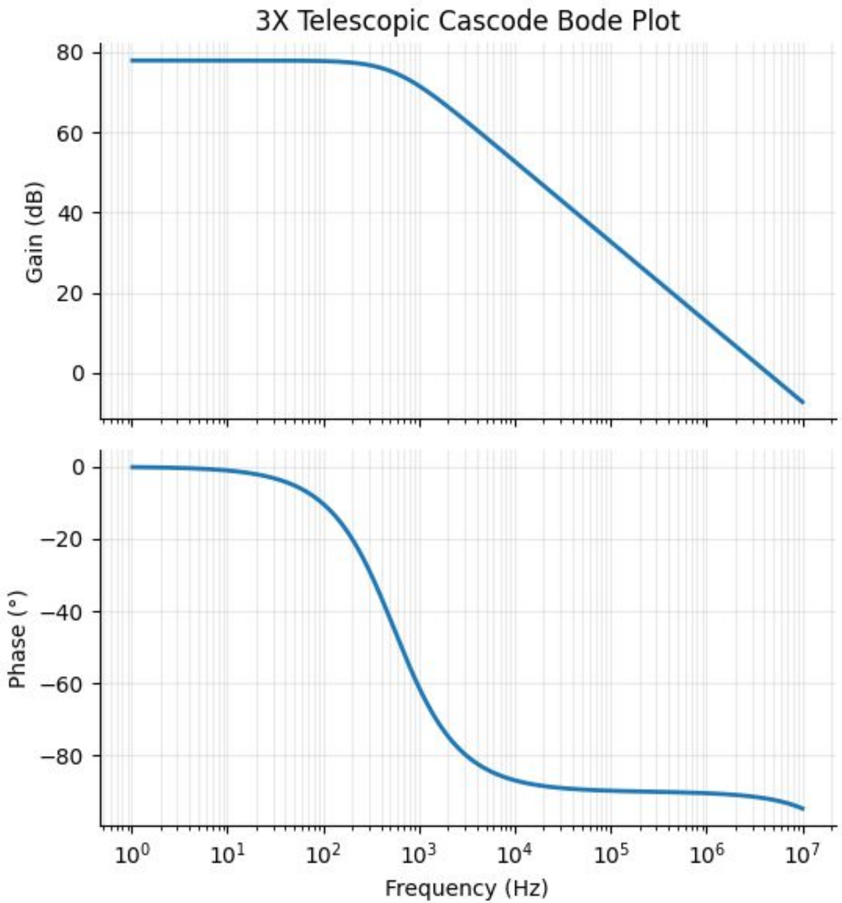
Gain = (76.04 dB)
PM = 83.9°
GBW = 5.75 MHz



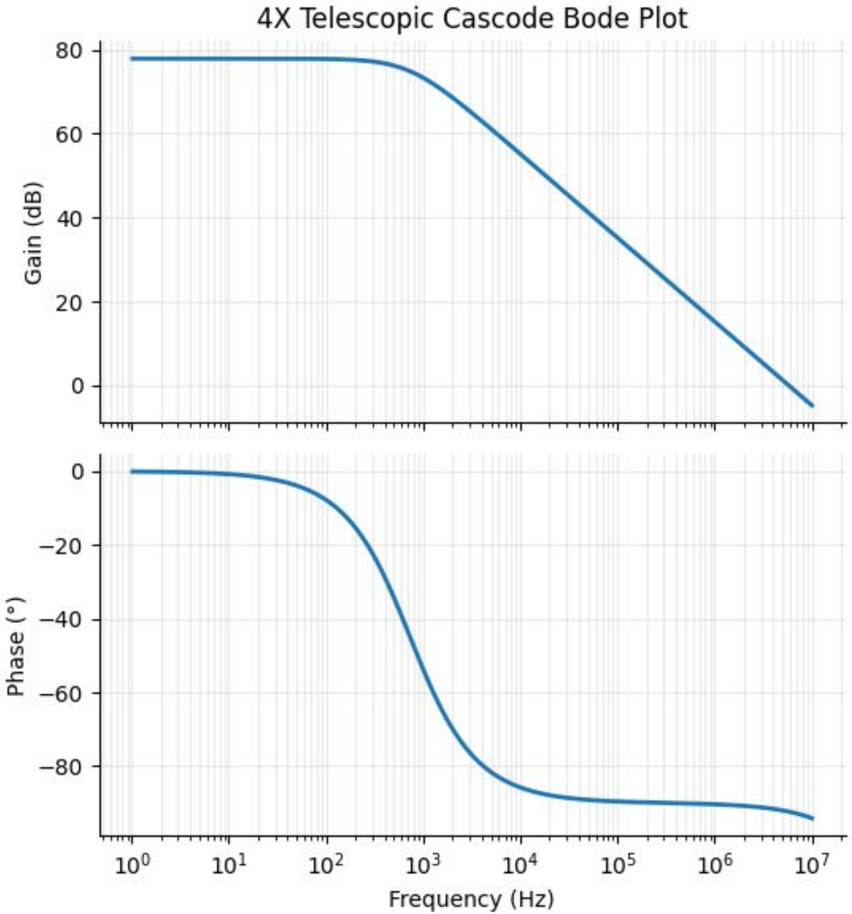
Gain = (77.90 dB)
PM = 89.2°
GBW = 1.43 MHz



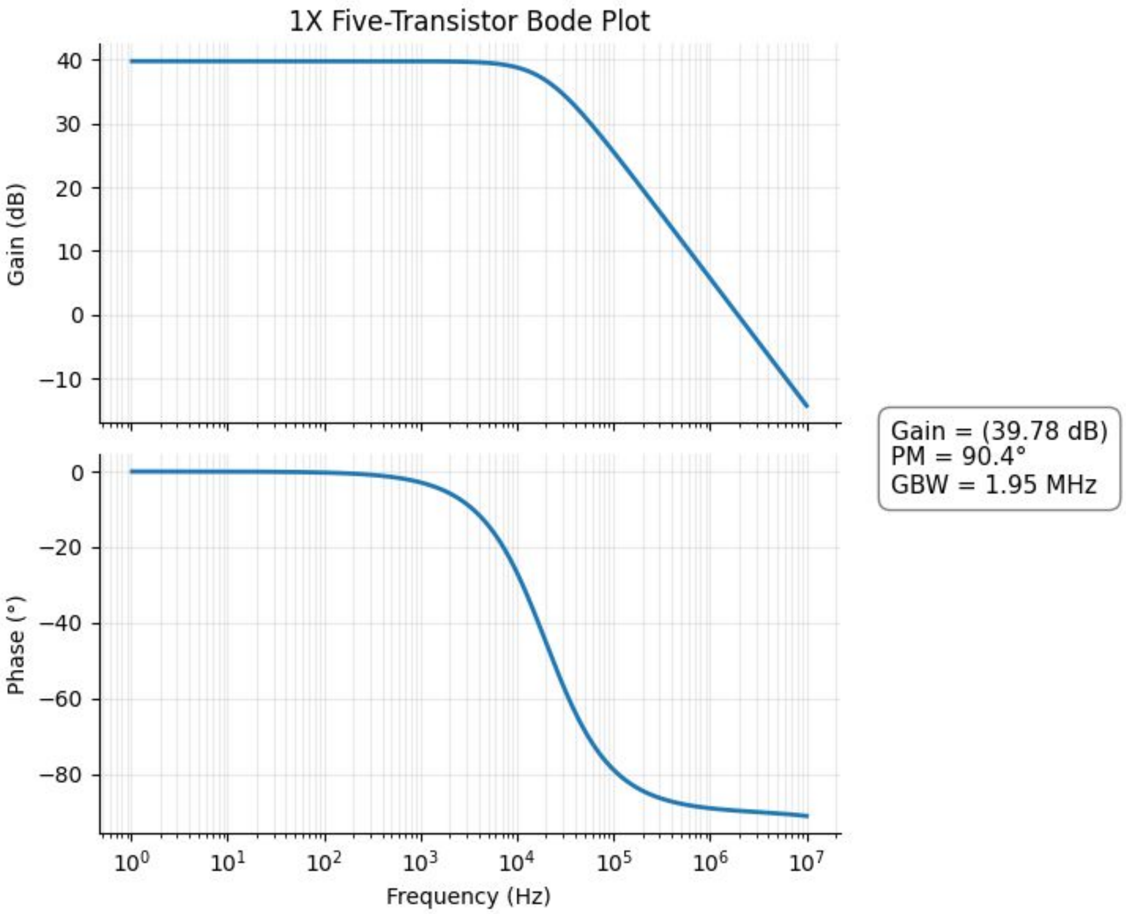
Gain = (77.87 dB)
PM = 88.8°
GBW = 2.88 MHz

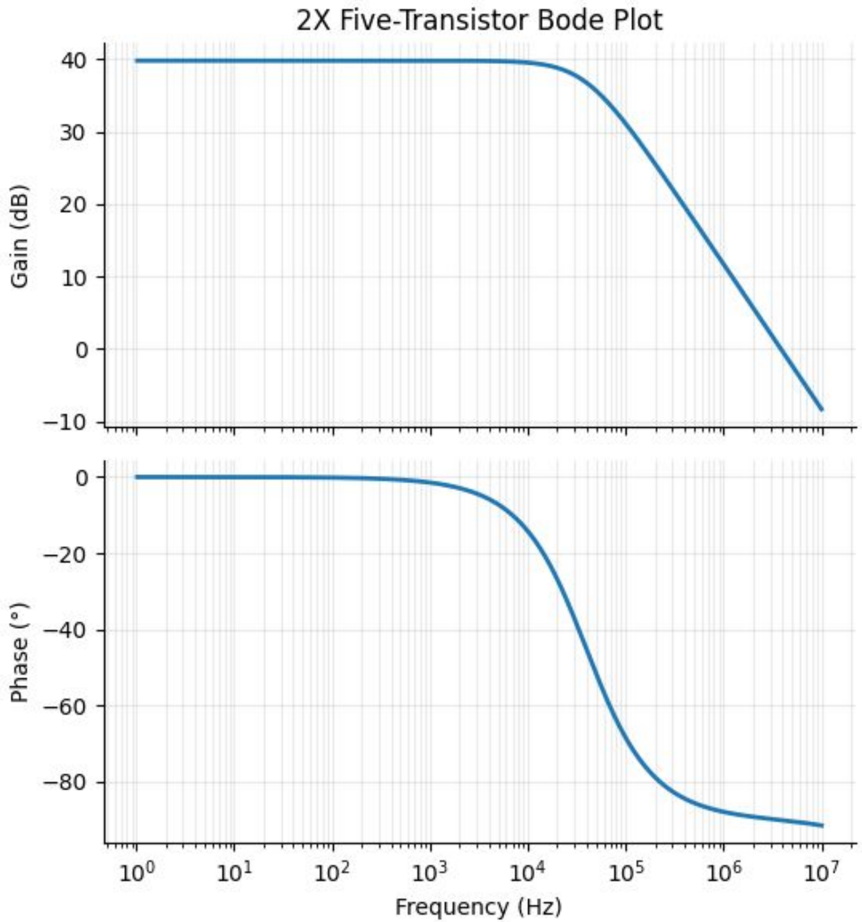


Gain = (77.87 dB)
PM = 88.0°
GBW = 4.37 MHz

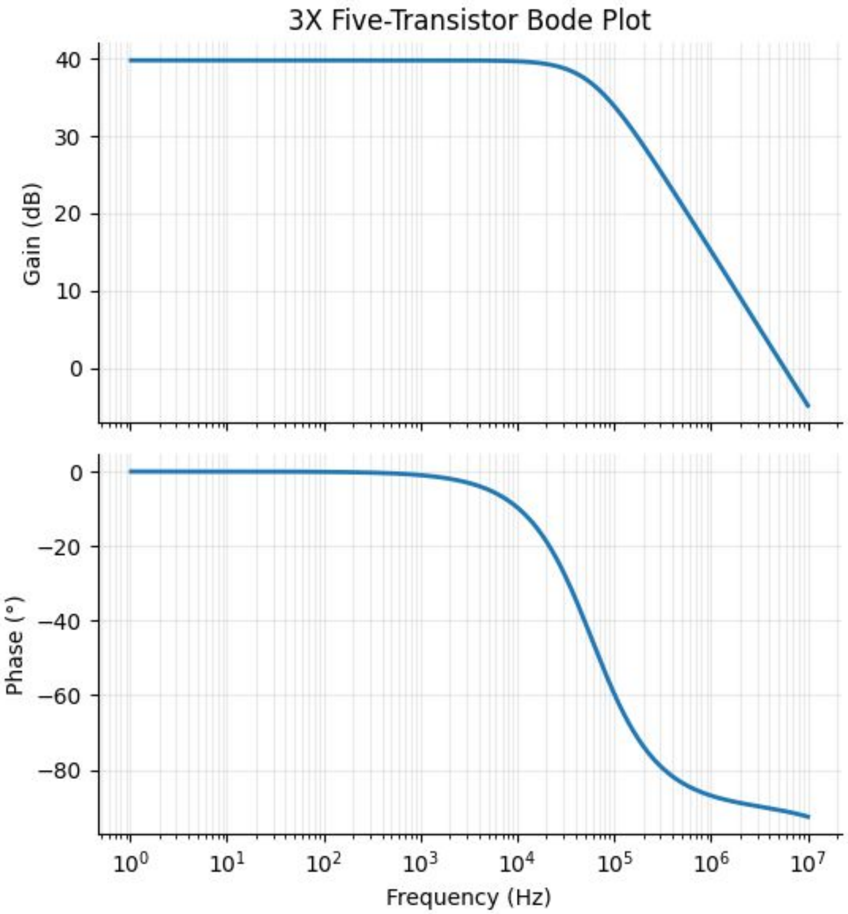


Gain = (77.87 dB)
PM = 87.7°
GBW = 5.75 MHz

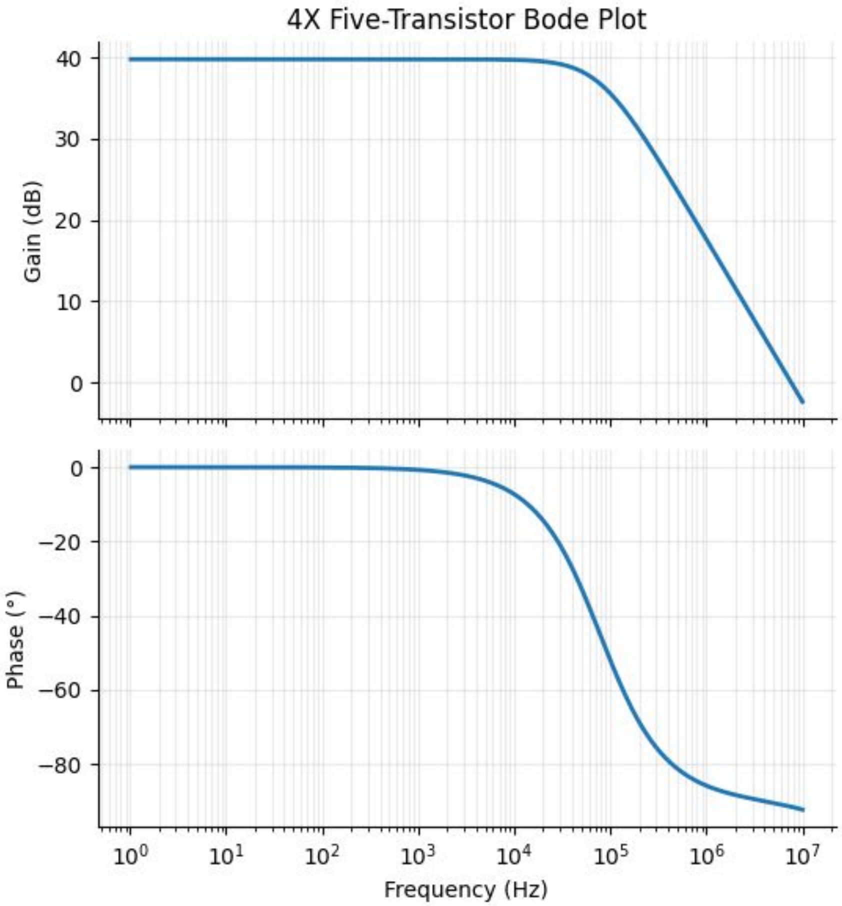

1X Five-Transistor Bode Plot
Gain (dB)
40
30
20
10
0
−10
Phase (°)
0
−20
−40
−60
−80
10^0
10^1
10^2
10^3
10^4
10^5
10^6
10^7
Frequency (Hz)
Gain = (39.78 dB)
PM = 90.4°
GBW = 1.95 MHz



Gain = (39.78 dB)
PM = 89.9°
GBW = 3.89 MHz



Gain = (39.78 dB)
PM = 88.9°
GBW = 5.75 MHz

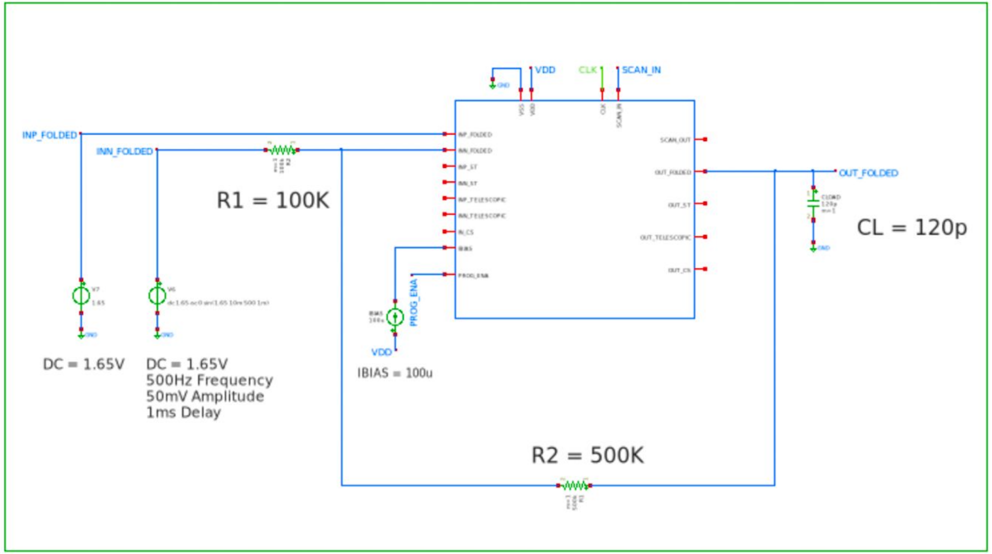


Gain = (39.78 dB)
PM = 88.5°
GBW = 7.76 MHz

Four Topology MOSbius as an Inverting Amplifier

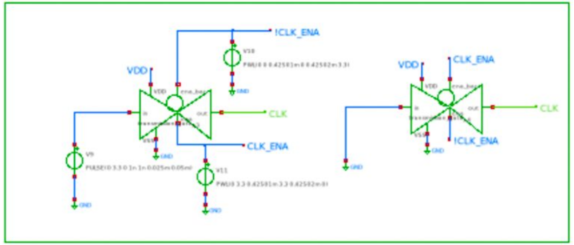
IMPLEMENTATION

Configure the folded cascode in 4× mode as an inverting amplifier to amplify a sine wave input by 5 while driving a 120pF capacitive load.



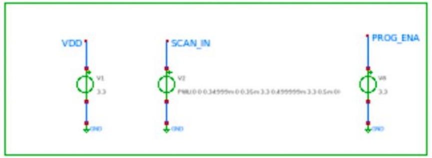
CLOCK GENERATION

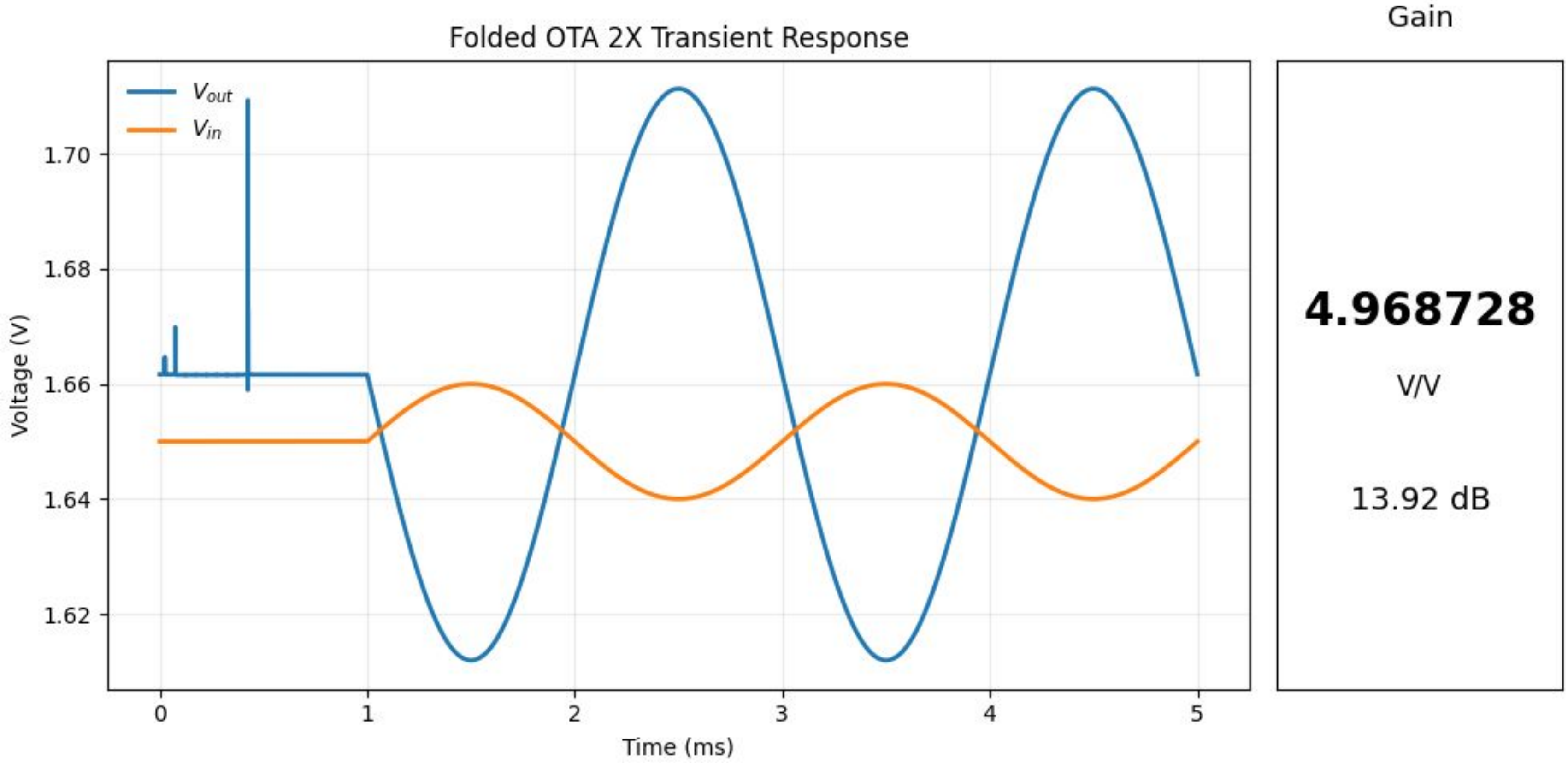
Artificially generate a clock signal and ensure it turns off after around .425ms, which is when the scan in signal has fully propogated through the scan chain



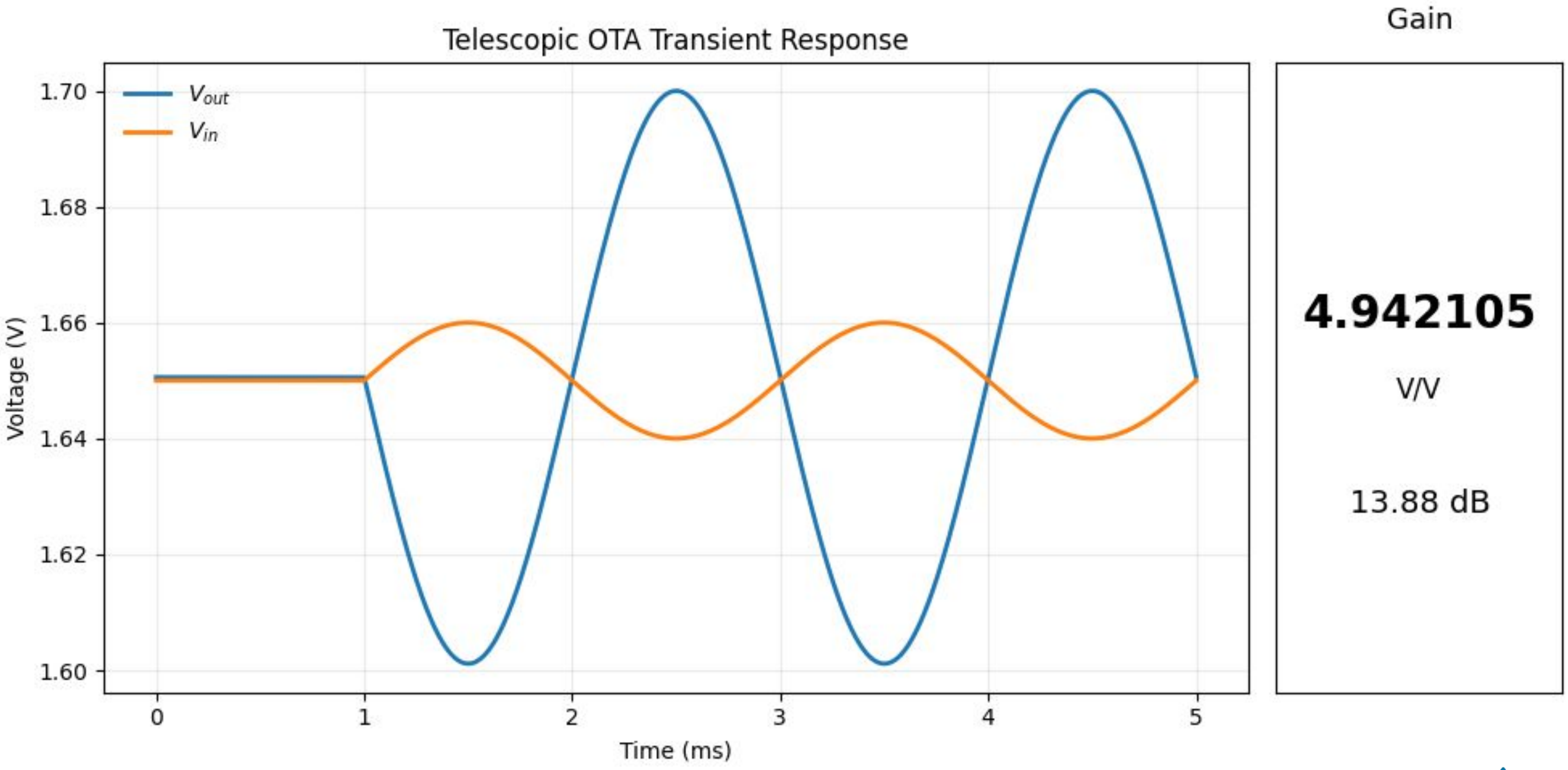
VDD, GND, SCAN INPUT, PROGRAMMABLE ENABLE

Scan chain input artificially generated to enable folded cascode 4x sizing
prog_ena sets default (low) or program (high)

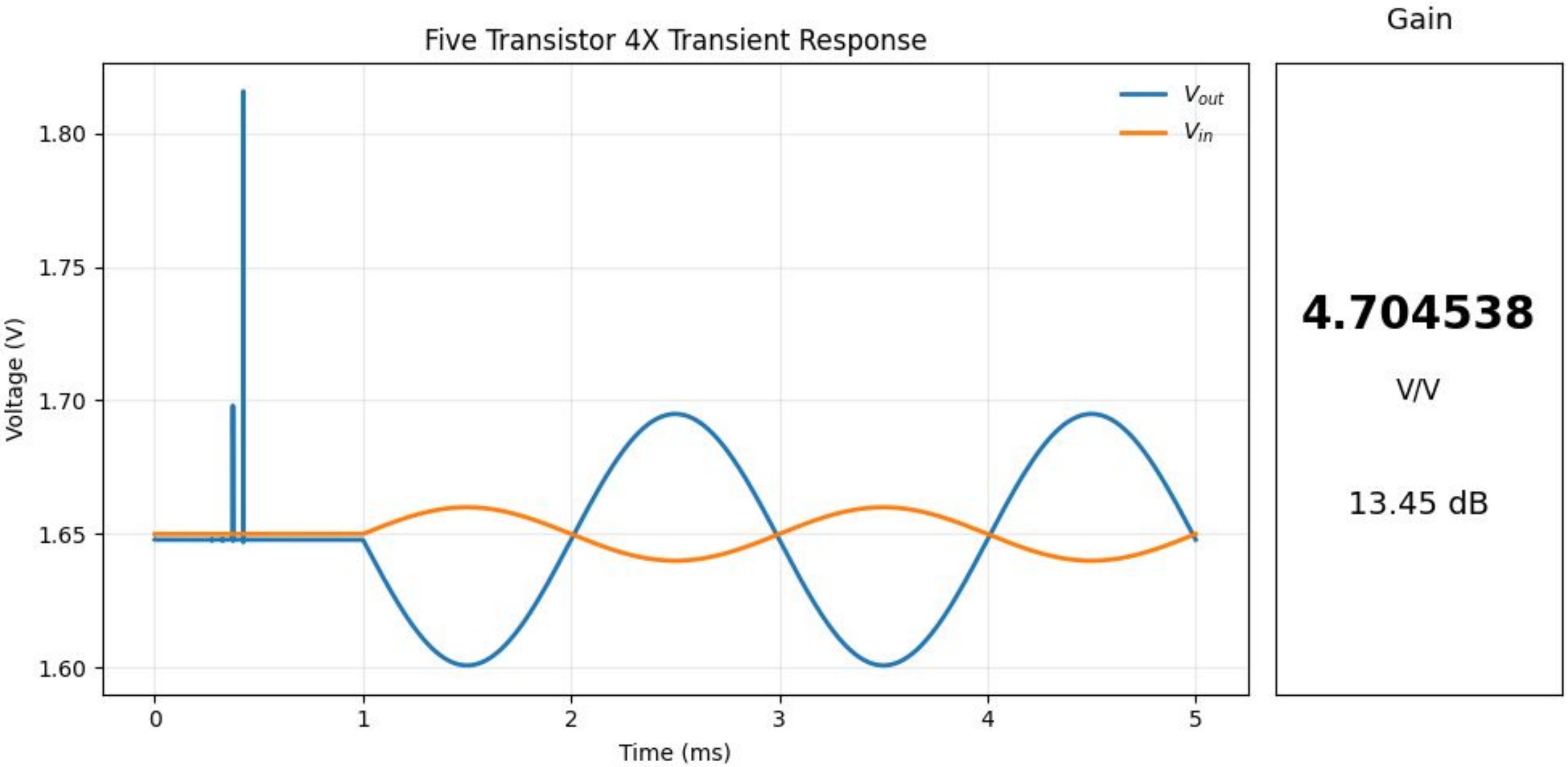




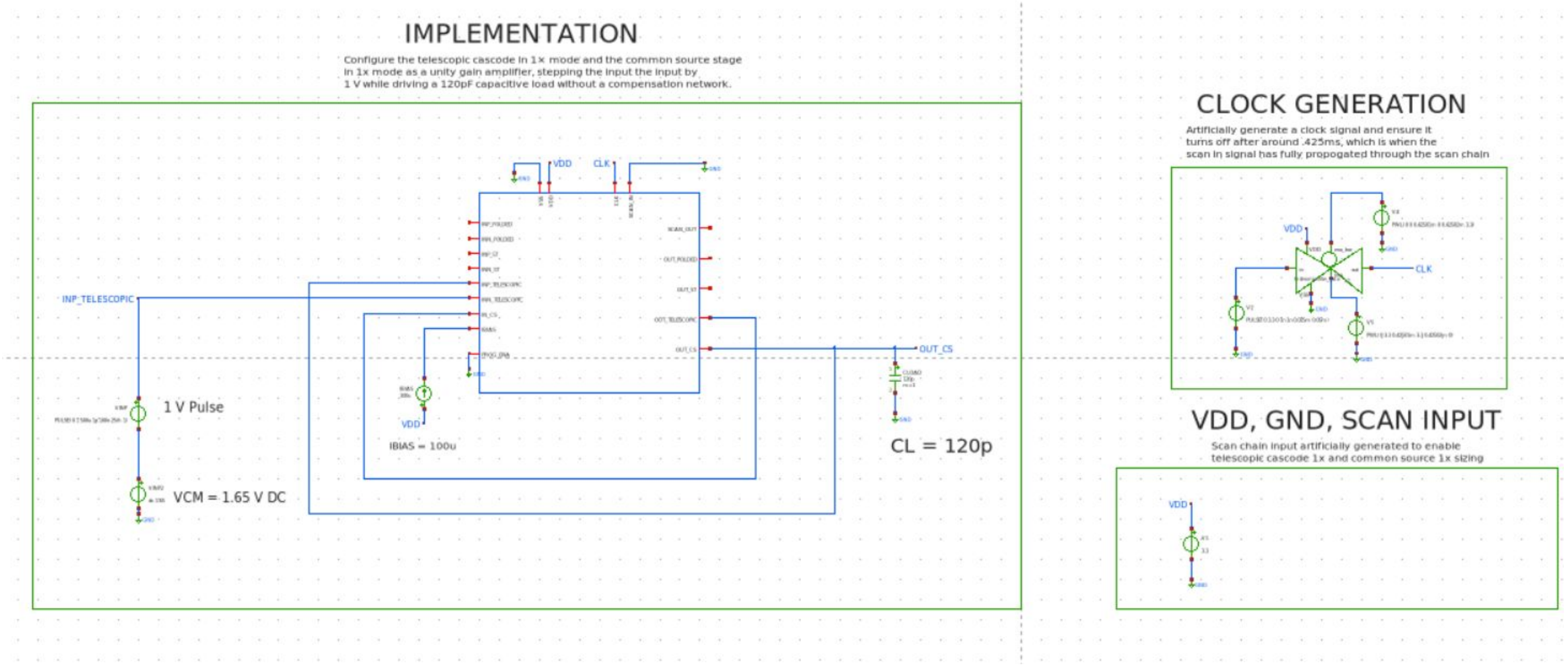
Folded OTA 2X Transient Response
V_{out}
V_{in}
1.70
1.68
1.66
1.64
1.62
Voltage (V)
0
1
2
3
4
5
Time (ms)
Gain
4.968728
V/V
13.92 dB

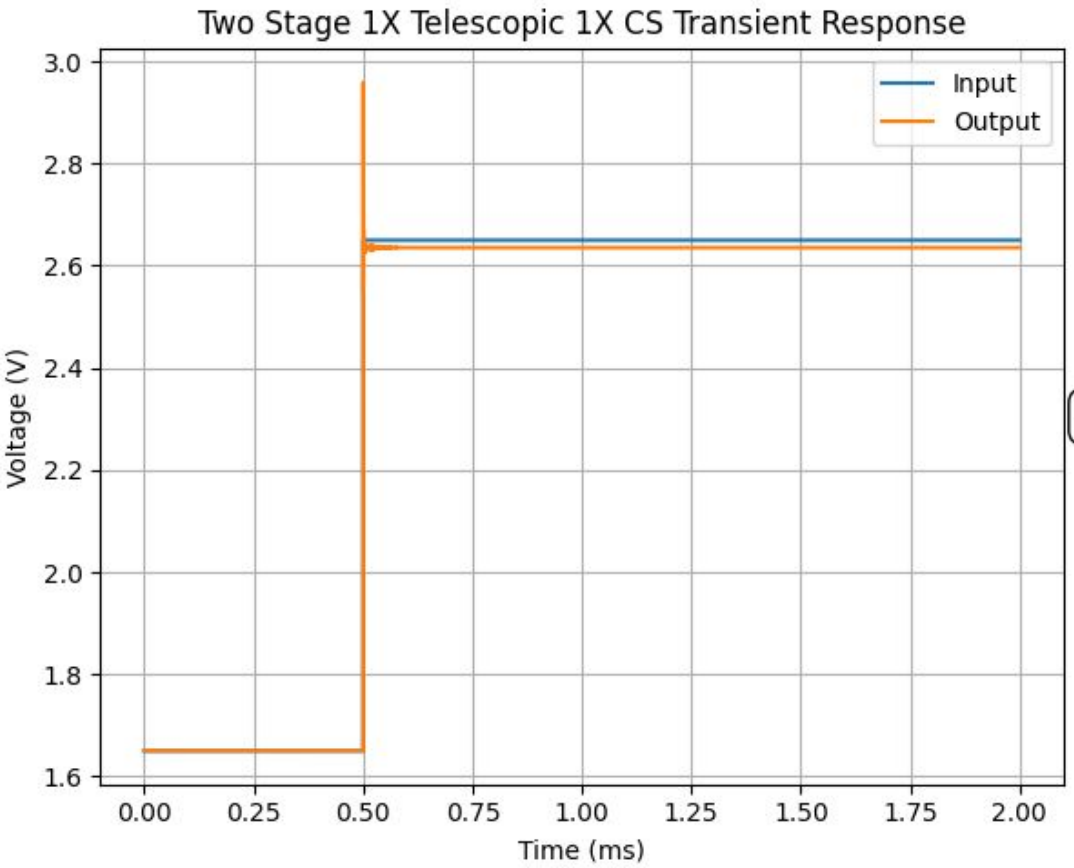


Telescopic OTA Transient Response
V_{out}
V_{in}
1.70
1.68
1.66
1.64
1.62
1.60
Voltage (V)
0
1
2
3
4
5
Time (ms)
Gain
4.942105
V/V
13.88 dB

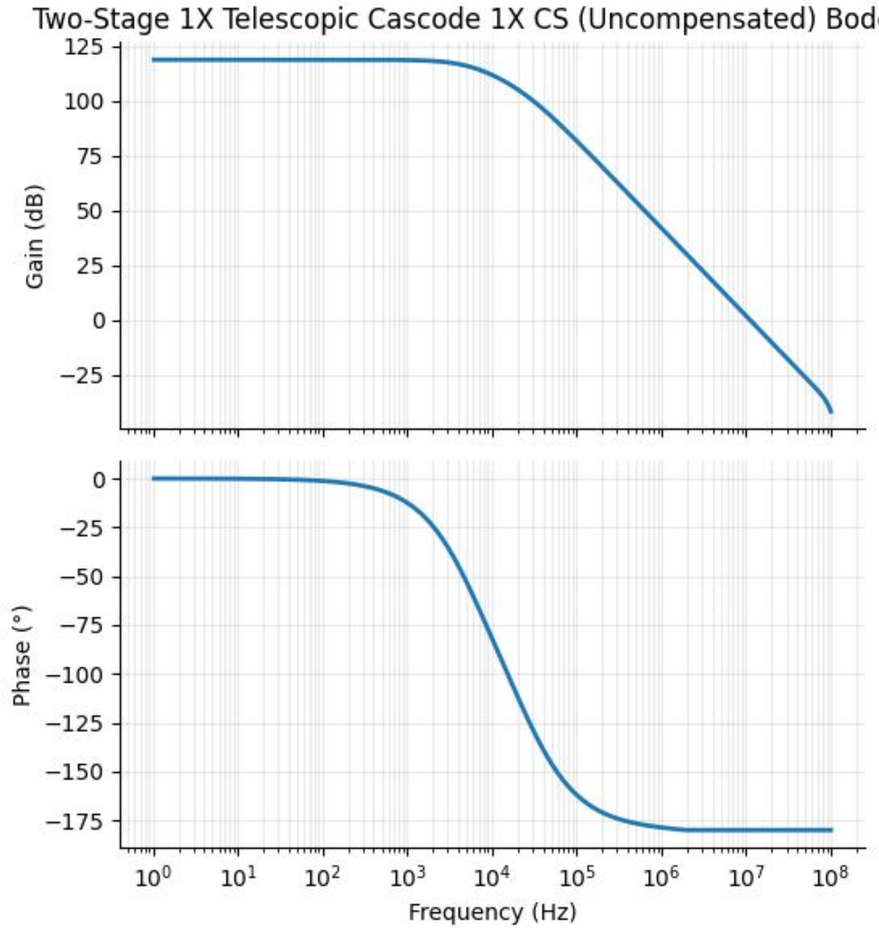

Five Transistor 4X Transient Response
Voltage (V)
1.80
1.75
1.70
1.65
1.60
0
1
2
3
4
5
Time (ms)
V_{out}
V_{in}
Gain
4.704538
V/V
13.45 dB

Four Topology MOSbius as a Two-Stage Amplifier


IMPLEMENTATION
Configure the telescopic cascode in 1x mode and the common source stage
in 1x mode as a unity gain amplifier, stepping the input the input by
1 V while driving a 120pF capacitive load without a compensation network.
VDD
CLK
INP_TELESCOPIC
1 V Pulse
VCM = 1.65 V DC
VDD
IBIAS = 100u
OUT_CS
CL = 120p
CLOCK GENERATION
Artificially generate a clock signal and ensure it
turns off after around .425ms, which is when the
scan in signal has fully propogated through the scan chain
VDD
CLK
VDD, GND, SCAN INPUT
Scan chain input artificially generated to enable
telescopic cascode 1x and common source 1x sizing
VDD



Phase Margin = 37.16°



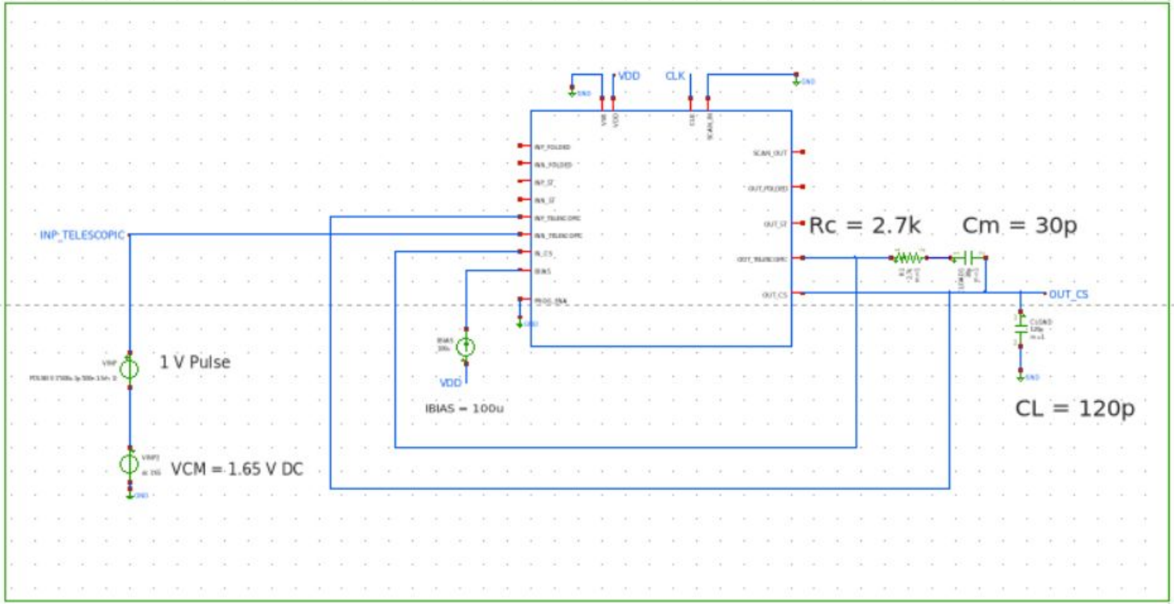
Gain = (118.62 dB)
PM = 0.0°
GBW = 11.22 MHz

Two-Stage Compensation

Users can also implement a Miller capacitor and compensation resistor to stabilize their amplifier configuration.

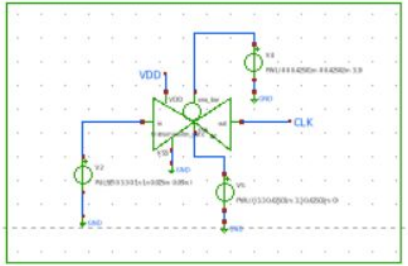
IMPLEMENTATION

Configure the telescopic cascode in 1x mode and the common source stage
in 1x mode as a unity gain amplifier, stepping the input the input by
1 V while driving a 120pF capacitive load without a compensation network.



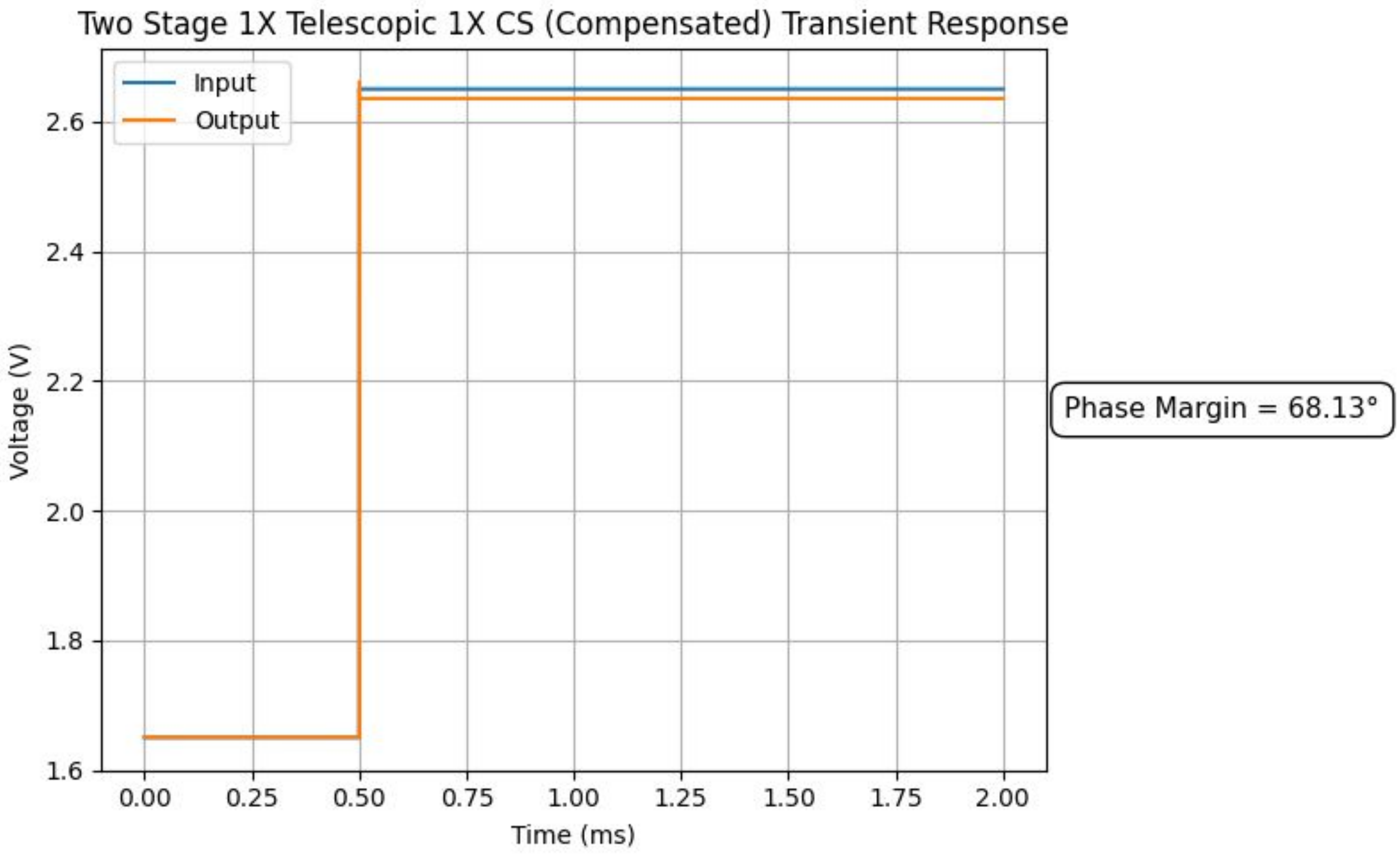
CLOCK GENERATION

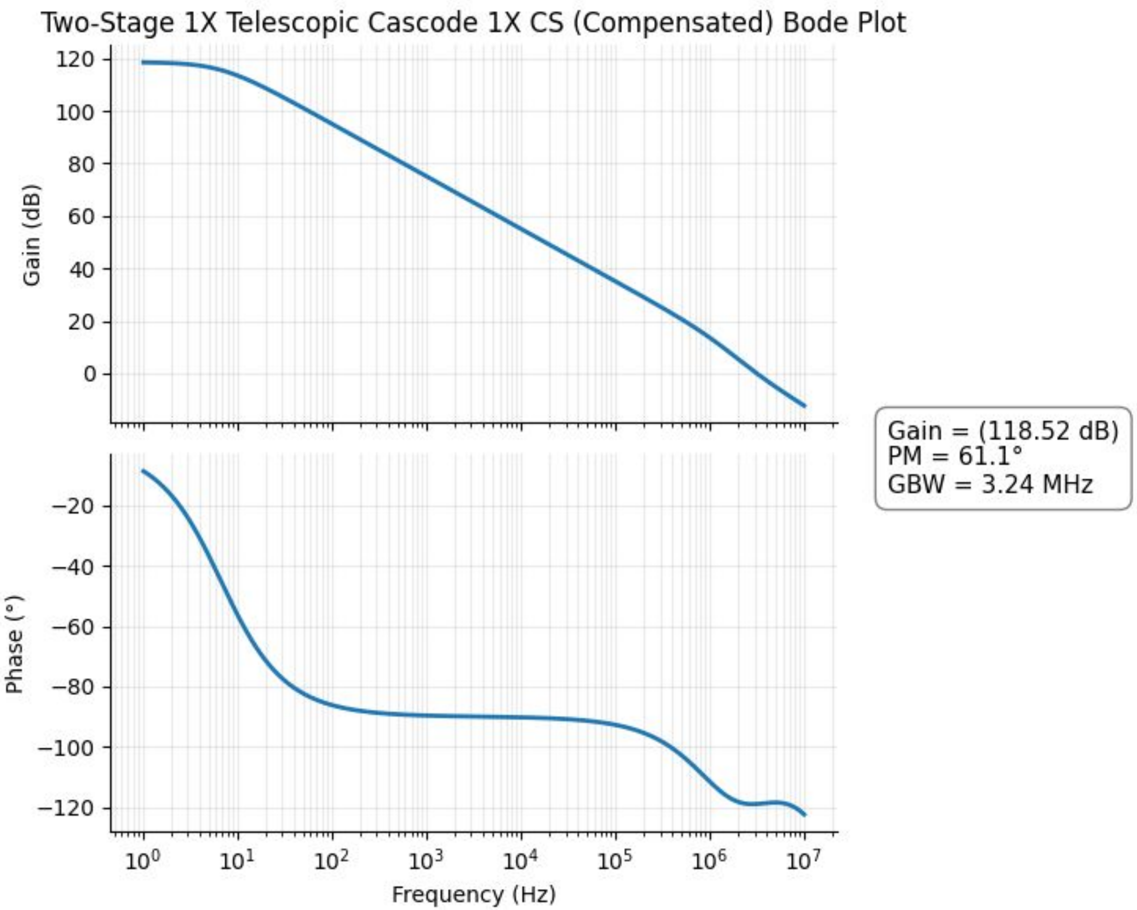
Artificially generate a clock signal and ensure it
turns off after around .425ms, which is when the
scan in signal has fully propogated through the scan chain



VDD, GND, SCAN INPUT

Scan chain input artificially generated to enable
telescopic cascode 1x and common source 1x sizing


Two Stage 1X Telescopic 1X CS (Compensated) Transient Response
Input
Output
Voltage (V)
2.6
2.4
2.2
2.0
1.8
1.6
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Time (ms)
Phase Margin = 68.13°


Two-Stage 1X Telescopic Cascode 1X CS (Compensated) Bode Plot
Gain (dB)
Phase (°)
Frequency (Hz)
Gain = (118.52 dB)
PM = 61.1°
GBW = 3.24 MHz

Full Verification Results:

Folded Cascode OTA:
https://docs.google.com/spreadsheets/d/1ste8NNTPVhHXfTPsn5K2P177Z7tUey5MOiy1HPbO30U/edit?usp=sharing

Telescopic Cascode OTA:
https://docs.google.com/spreadsheets/d/1UWE3XYKk2p_d3-MaK_v5p6VSCiXoplRrxV4e-1WU_l0/edit?usp=sharing

5-Transistor OTA:
https://docs.google.com/spreadsheets/d/12NohRTnEUuWv6hk_VIJNx9IA-sGq2unuuVgBkl8fErU/edit?usp=sharing